

PYTHON I 12 - 16

Lesson 15. Calculator

Lesson description

Students will create a calculator using the Tkinter library. In this lesson, they will put the widgets (buttons, entry fields) and also program those buttons until they work like a calculator program.

Tools:

1. PyCharm

Learning Objectives

Students will:

1. design the calculator
2. implement their knowledge about placement
3. create a function for each button
4. create a function for entry fields

Timing	Lesson description	Additional materials	Comments
5 min	Preparing • Guide students to prepare the project directory. • Tell students to open Pycharm • Make sure Internet Connection is Good • Review the lesson		

5 min	Comprehension check • We ask students to write the things they have learned during the last lesson on the board. • We ask students about the syntax we used in the last lesson. For example, we can ask students if they still remember how to create a button, a label, and an entry field.		
5 min	Flow of the Project • Teacher shows the result of the program and explains the flow of the program • The first step to creating the program is to design the application widget • After we design the calculator, we will program each button to calculate the result.		
5 min	Create a window • Import the Tkinter Library before creating the window. • Guide the students to create a new window to put your widget on it. • Change the window size geometry into (250x300)		
35 min	Widget • Create a variable for every widget on Pycharm. For example, (Button1, Button2, etc.) • Guide students to place every widget that we will create (Button, Entry Fields) • change the size of the widget that we put on the window • Entry Fields: this widget will have a special variable (Text Variable):		

	○ Input Text is a special variable that we define to manipulate the value of Entry Fields that we get from buttons.		
30 min	Functions Explain and Code ● Guide the students to create a function that triggered the Button and Entry Fields. ● Lambda Function: Tell the students that we will use the Lambda function when we call the function on the button to pass the value into Entry Fields ● in this program, we will have three functions: ○ Insert_Number is a function to pass the value into entry fields. ○ Button_Clear is a function to delete the value in Entry Fields ○ Result is a function to see the result of the value calculations. ▪ Expression is a variable to gather the data from the Button before we pass the value to Entry Fields ▪ Eval(Expression) is a function when we collect the data in the Entry Fields. Those data will be calculated based on the symbols you have on the button.		
5	Review ● Review the lesson ● checking students' design		

